

WEEK 2 | PRACTICAL 1

Give every value a meaningful type • fixed-value typed profiles

Today’s boundary

Use fixed values only. No Scanner, if/else, loops, methods beyond main, arrays, files, locker assignment, pickup codes, fees or analytics.

Learning outcomes		
Outcome	Student evidence	Status
Classify field meaning	Text, whole number, decimal, character or true/false	Done
Choose legal identifiers and literals	Names and literal forms are explained	Done
Use primitive types and String correctly	Both application dictionaries are complete	Done
Use final named constants	Rules are not magic numbers	Done
Compile, run and diagnose	Prediction, output and one repaired fault	Done

U1 / S1 — classify the two application records						
Application	Field	Example	Category	Chosen Java type	Why this type?	Status
UniFlow	studentId	U26A001	Text	string	It stores value of letters and numbers	Done
UniFlow	studentName	Asha Rao	Text	string	It stores letters	Done
UniFlow	department	CSE	Text	string	it stores letters	Done
UniFlow	semester	1	numeric	int	it stores integer values	Done
UniFlow	attendancePercent	82.5	numeric	float	it stores decimal values	Done
UniFlow	sectionCode	A	Text	char	it stores single character in single quotes	Done
UniFlow	feePaid	TRUE	logical/text	boolean	it stores the value of TRUE or FALSE	Done
UniFlow	applicationNumber	2026000001	numeric	long	it stores very long integers	Done
Smart-Locker	parcelId	P-1001	Text	string	It stores value of letters and numbers	Done
Smart-Locker	recipientName	Ravi Kumar	Text	string	It stores letters	Done
Smart-Locker	sizeCode	M	text	char	it stores single character in single quotes	Done
Smart-Locker	weightKg	2.75	numeric	float	it stores decimal values	Done

Smart-Locker	accepted	TRUE	logical/text	boolean	it stores the value of TRUE or FALSE	Done
Smart-Locker	intakeSequence	1001	numeric	int	it stores integer values	Done
Smart-Locker	receivedEpochSeconds	1786406400	numeric	Long	it stores very long integers	Done

U2 / S2 — constants and literal quick check

Rule / literal	Your answer	Reason
MAX_SEMESTERS = 8	variable or final constant?	it is final constant because the variable is written is all caps it mean the value locked
MIN_ATTENDANCE_PERCENT = 75.0	variable or final constant?	it is also final constant because the variable is written is all caps it mean the value locked
26001 vs 26001L	int or long?	26001 is integer because it holds normal integer value and 26001L is long integer while is mentioned at
2.75 vs 2.75F	double or float?	2.75 is double because it hole some acurate value and 2.75F is float because the value is
'M' vs "M"	char or String?	char is single letter sotred in single quotes and string stores multiple words in double quotes
true	boolean meaning	it mean it stores the TRUE or FALSE values only

U3 / S2 — compile and run evidence

File	Compile command	Run command	Predicted output	Actual output	First compiler message / none	Status
UniFlowTypedProfile.java	javac UniFlowTypedProfile.java	java UniFlowTypedProfile	UniFlow Typed Profile Student ID: U26A001 Student name: Asha Rao Department: CSE Semester: 1 Attendance percent: 82.5 Section code: A Fee paid: true Application number: 2026000001 Maximum semesters: 8 Minimum attendance percent: 75.0	UniFlow Typed Profile Student ID: U26A001 Student name: Asha Rao Department: CSE Semester: 1 Attendance percent: 82.5 Section code: A Fee paid: true Application number: 2026000001 Maximum semesters: 8 Minimum attendance percent: 75.0	none	Done
SmartLockerTypedParcel.java	javac SmartLockerTypedParcel.java	java SmartLockerTypedParcel	Smart-Locker Typed Parcel Parcel ID: P-1001 Recipient name: Ravi Kumar Size code: M Weight (kg): 2.75 Accepted: true Intake sequence: 1001 Received epoch seconds: 1786406400 Minimum parcel weight (kg): 0.1 Maximum parcel weight (kg): 20.0	Smart-Locker Typed Parcel Parcel ID: P-1001 Recipient name: Ravi Kumar Size code: M Weight (kg): 2.75 Accepted: true Intake sequence: 1001 Received epoch seconds: 1786406400 Minimum parcel weight (kg): 0.1 Maximum parcel weight (kg): 20.0	none	Done

Type-error clinic

Fault	Single change to test	Prediction	Observed message	Plain-language diagnosis	Smallest repair	Status
F1	double weightKg -> int while keeping 2.75	compiler throws error	Type mismatch: cannot convert from double to int	in the programe the comiler cant store decimal values in int so this mismatch causes error	removing 'int' and adding 'double'	Done

F2	char sizeCode = "M"	compiler throws error	Type mismatch: cannot convert from String to char	here the using double quotes makes the stored value string and decleration of char mismatches here and throws error	removing double qoutes and placing single quote beside 'M'	Done
F3	float temperatureC = 24.5	compiler throws error	Type mismatch: cannot convert from double to Float	the compiler throwed error because the declared datatype is Float and stored value in it is 'double'.	adding 'f' to end of value, makes the value float ones	Done
F4	reassign a final constant	compiler throws error	output came	no error occured	NA	Done

P1 completion check

Check	Done?	Evidence path / note
Both data dictionaries explain field meaning and type	Done	C:\Documents\PSPJ
Both Java files compile and print labelled output	Done	C:\Documents\PSPJ
One peer challenged a type choice	Done	C:\Documents\PSPJ
One fault was predicted, repaired and re-tested	Done	C:\Documents\PSPJ

Runnable programs and expected output

Run both programs before completing the evidence tables. Copy each complete code cell into a file with the exact filename, compile it, run it, and explain every output line.

Program 1 — UniFlowTypedProfile.java

```
public class UniFlowTypedProfile {
    public static void main(String[] args) {
        final int MAX_SEMESTERS = 8;
        final double MIN_ATTENDANCE_PERCENT = 75.0;

        String studentId = "U26A001";
        String studentName = "Asha Rao";
        String department = "CSE";
        int semester = 1;
        double attendancePercent = 82.5;
        char sectionCode = 'A';
        boolean feePaid = true;
        long applicationNumber = 2026000001L;

        System.out.println("=== UniFlow Typed Profile ===");
        System.out.println("Student ID: " + studentId);
        System.out.println("Student name: " + studentName);
        System.out.println("Department: " + department);
        System.out.println("Semester: " + semester);
        System.out.println("Attendance percent: " + attendancePercent);
        System.out.println("Section code: " + sectionCode);
        System.out.println("Fee paid: " + feePaid);
        System.out.println("Application number: " + applicationNumber);
        System.out.println("Maximum semesters: " + MAX_SEMESTERS);
        System.out.println("Minimum attendance percent: " + MIN_ATTENDANCE_PERCENT);
    }
}
```

Expected output — UniFlowTypedProfile

```
=== UniFlow Typed Profile ===
Student ID: U26A001
Student name: Asha Rao
Department: CSE
Semester: 1
Attendance percent: 82.5
Section code: A
Fee paid: true
Application number: 2026000001
Maximum semesters: 8
Minimum attendance percent: 75.0
```

Program 2 — SmartLockerTypedParcel.java

```
public class SmartLockerTypedParcel {
    public static void main(String[] args) {
        final double MIN_PARCEL_WEIGHT_KG = 0.1;
        final double MAX_PARCEL_WEIGHT_KG = 20.0;

        String parcelId = "P-1001";
        String recipientName = "Ravi Kumar";
        char sizeCode = 'M';
        double weightKg = 2.75;
        boolean accepted = true;
        int intakeSequence = 1001;
        long receivedEpochSeconds = 1786406400L;

        System.out.println("=== Smart-Locker Typed Parcel ===");
        System.out.println("Parcel ID: " + parcelId);
        System.out.println("Recipient name: " + recipientName);
        System.out.println("Size code: " + sizeCode);
        System.out.println("Weight (kg): " + weightKg);
        System.out.println("Accepted: " + accepted);
        System.out.println("Intake sequence: " + intakeSequence);
        System.out.println("Received epoch seconds: " + receivedEpochSeconds);
        System.out.println("Minimum parcel weight (kg): " + MIN_PARCEL_WEIGHT_KG);
        System.out.println("Maximum parcel weight (kg): " + MAX_PARCEL_WEIGHT_KG);
    }
}
```

Expected output — SmartLockerTypedParcel

```
=== Smart-Locker Typed Parcel ===
Parcel ID: P-1001
Recipient name: Ravi Kumar
Size code: M
Weight (kg): 2.75
Accepted: true
Intake sequence: 1001
Received epoch seconds: 1786406400
Minimum parcel weight (kg): 0.1
Maximum parcel weight (kg): 20.0
```

Compile and run: `javac UniFlowTypedProfile.java → java UniFlowTypedProfile; javac SmartLockerTypedParcel.java → java SmartLockerTypedParcel`. Paste your actual console output into the U3 table above.

If you change a value, update your prediction before running. Keep the final `.java` files compiling; record broken experiments only in the fault log or a text file.

Submission reminder: include both `.java` files, dictionaries, evidence and Git history. Do not submit generated `.class` files.